

Concordia University

Dept. of Computer Science & Software Engineering
COMP 353 – Databases — Summer 2026

Warm-up Project Report

A Simple Database Application for the Country Soccer Club System (CSCS)

Group Account: wqc353_1

Due: July 26, 2026

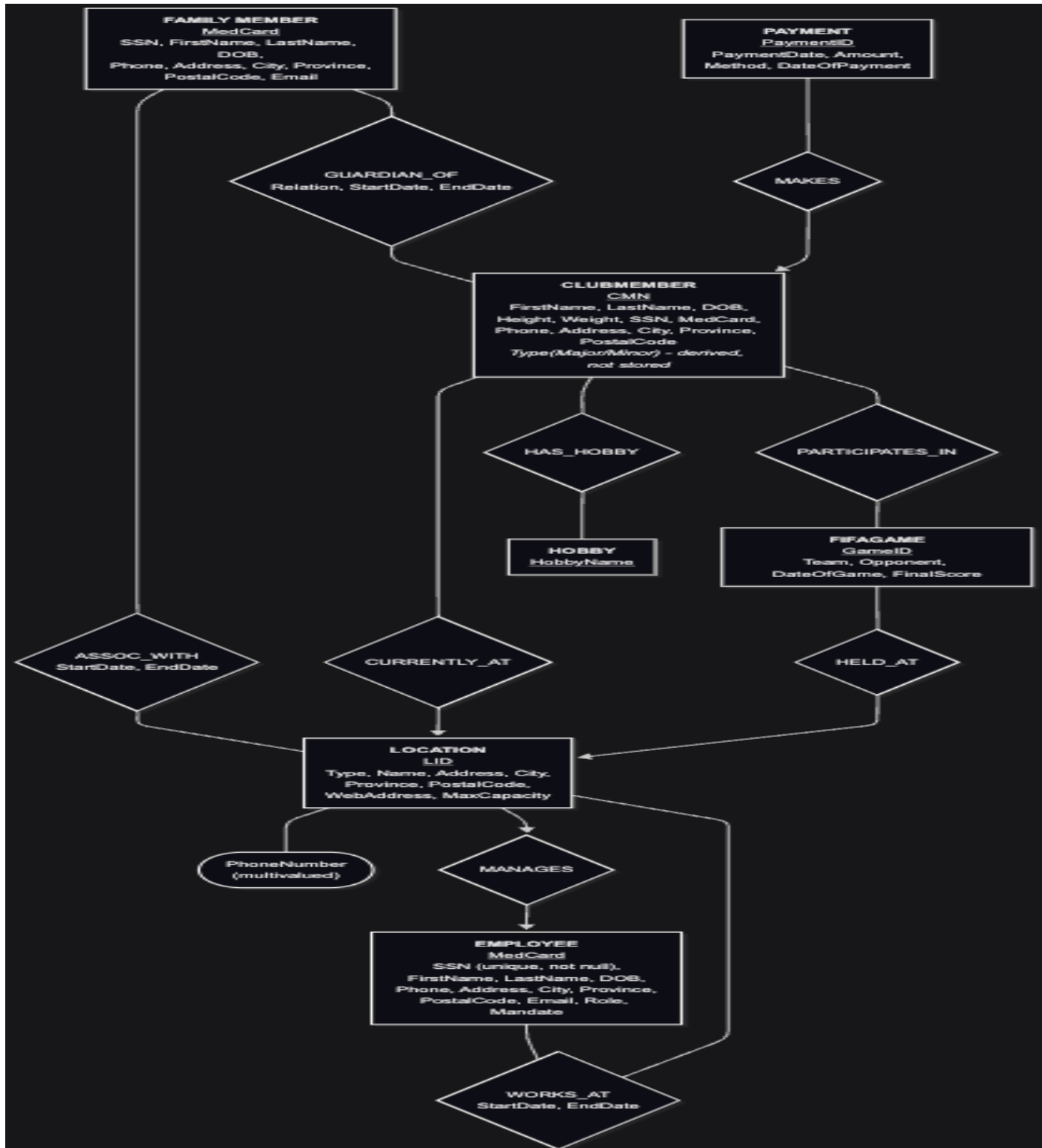
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Repository: [cscs-db-353](#)

1. Design — E/R Diagram

The Country Soccer Club System (CSCS) is modeled above uses the *Entity-Relationship model*.



LEGEND

- Rectangles represent entity sets;
- Diamonds represent **relationship sets**;
- Ovals represent **multivalued attributes**;
- **Primary key attributes are marked (PK)**;
- Arrowheads point to the "1" side of a relationship; and
- Plain line marks the "many" side.

Key Design Decisions

- The Personnel-to-Location and FamilyMember-to-Location associations are time-dependent (WORKS_AT, ASSOC_WITH); thus, each has its own associative table that has its own StartDate/EndDate attributes, as opposed to a foreign key column on the entity. That is because a person may change locations and all of this needs to be tracked.
- The association between ClubMember and FamilyMember (GUARDIAN_OF) is time-dependent and is M:N – that is, a minor club member can have different family members as his guardians at different periods of time, and one and the same family member can be the guardian of multiple club members.
- The FIFAGAME entity is a strong entity which has M:N association with the ClubMember entity through the PARTICIPATES_IN relationship. In this way, we can avoid repetition of the game information (opponent, score, date) for each individual participant
- ClubMember.Type (Major/Minor) and "active/inactive" attribute (membership status) are derived attributes, computed on demand based on DateOfBirth and payment history attributes, respectively. We do not want to store those in the database; otherwise, the data will get out-of-date.
- The CMN (membership number) attribute is a single, system-wide auto-increment attribute serving as a primary key.

2. Database Schema — DDL, Keys, and Relationships

The above E/R diagram has been transformed into relational forms using the standard transformation rules that include: An entity set is transformed to a table; A many-to-many or temporal relationship is transformed into a junction table which has either a composite key or surrogate key; A one-to-many relationship is transformed into a foreign key at the “many” end; and

2.1 Primary Keys

Table	Primary Key
Location	location_id
LocationPhone	(location_id, phone_number)
Personnel	personnel_id
PersonnelAssignment	assignment_id
FamilyMember	family_member_id
FamilyMemberAssignment	assignment_id
ClubMember	membership_number (AUTO_INCREMENT)
ClubMemberFamilyRelation	relation_id
Hobby	hobby_id
ClubMemberHobby	(membership_number, hobby_id)
Payment	payment_id
FIFAGame	game_id
FIFAParticipation	(game_id, membership_number)

2.2 Foreign Keys

Table	Foreign Key	References
PersonnelAssignment	personnel_id	Personnel
PersonnelAssignment	location_id	Location
FamilyMemberAssignment	family_member_id	FamilyMember
FamilyMemberAssignment	location_id	Location
ClubMember	location_id	Location
ClubMemberFamilyRelation	membership_number	ClubMember
ClubMemberFamilyRelation	family_member_id	FamilyMember
ClubMemberHobby	membership_number	ClubMember
ClubMemberHobby	hobby_id	Hobby
Payment	membership_number	ClubMember
FIFAParticipation	membership_number	ClubMember
FIFAParticipation	game_id	FIFAGame
LocationPhone	location_id	Location

2.3 Constraints

- Personnel.ssn and FamilyMember.ssn are NOT NULL and UNIQUE, per the specification.
- Medicare numbers are UNIQUE for Personnel and FamilyMember (no two people share a Medicare card number).
- Personnel role is restricted to an ENUM of the roles named in the specification (General Manager, Deputy Manager, Treasurer, Secretary, Administrator, Captain, Coach, Assistant Coach, Other).
- Personnel mandate is restricted to ENUM('Volunteer', 'Salaried').
- Payment method is restricted to ENUM('Cash', 'Debit', 'Credit').
- ClubMember.membership_number is a global AUTO_INCREMENT primary key, unique across all locations.
- Foreign keys ensure referential integrity everywhere — no orphaned rows for assignments, payments, hobbies or game participation.
- DATE type is used for all date columns; DECIMAL(10,2) type is used for money columns; DECIMAL(5,2) type is used for height/weight columns.

2.4 Data Types Used

Data Type	Purpose
INT	Primary and foreign keys
VARCHAR	Names, addresses, emails, phone numbers, SSN, Medicare numbers
DATE	Birth dates, assignment dates, payment dates, game dates
DECIMAL(10,2)	Payment amounts
DECIMAL(5,2)	Height and weight
ENUM	Personnel role, mandate, payment method, location type, relationship type

2.5 Full Schema DDL (01_schema.sql)

```
-- Country Soccer Club System

USE wqc353_1;

CREATE TABLE Location (
    location_id INT AUTO_INCREMENT PRIMARY KEY,
    location_type ENUM('Head', 'Branch') NOT NULL,
    name VARCHAR(100) NOT NULL UNIQUE,
    address VARCHAR(150) NOT NULL,
    city VARCHAR(60) NOT NULL,
    province CHAR(2) NOT NULL,
    postal_code VARCHAR(10) NOT NULL,
    web_address VARCHAR(200),
    capacity INT NOT NULL
);

CREATE TABLE LocationPhone (
    location_id INT NOT NULL,
    phone_number VARCHAR(20) NOT NULL,
    PRIMARY KEY (location_id, phone_number),
    FOREIGN KEY (location_id) REFERENCES Location(location_id)
    ON DELETE CASCADE
);

CREATE TABLE Personnel (
    personnel_id INT AUTO_INCREMENT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    date_of_birth DATE NOT NULL,
    ssn VARCHAR(15) NOT NULL UNIQUE,
    medicare_number VARCHAR(20) UNIQUE,
    phone_number VARCHAR(20),
    address VARCHAR(150),
    city VARCHAR(60),
    province CHAR(2),
    postal_code VARCHAR(10),
    email VARCHAR(100),
    role ENUM(
        'General Manager',
        'Deputy Manager',
        'Treasurer',
        'Secretary',
        'Administrator',
        'Captain',
        'Coach',
        'Assistant Coach',
        'Other'
    ) NOT NULL,
    mandate ENUM('Volunteer', 'Salaried') NOT NULL
);

CREATE TABLE PersonnelAssignment (
    assignment_id INT AUTO_INCREMENT PRIMARY KEY,
    personnel_id INT NOT NULL,
    location_id INT NOT NULL,
    start_date DATE NOT NULL,
    end_date DATE,
    UNIQUE (personnel_id, location_id, start_date),
    FOREIGN KEY (personnel_id) REFERENCES Personnel(personnel_id),
    FOREIGN KEY (location_id) REFERENCES Location(location_id),
    CHECK (end_date IS NULL OR end_date >= start_date)
);
```

```

CREATE TABLE FamilyMember (
    family_member_id INT AUTO_INCREMENT PRIMARY KEY,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    date_of_birth DATE NOT NULL,
    ssn VARCHAR(15) NOT NULL UNIQUE,
    medicare_number VARCHAR(20) UNIQUE,
    phone_number VARCHAR(20),
    address VARCHAR(150),
    city VARCHAR(60),
    province CHAR(2),
    postal_code VARCHAR(10),
    email VARCHAR(100)
);

CREATE TABLE FamilyMemberAssignment (
    assignment_id INT AUTO_INCREMENT PRIMARY KEY,
    family_member_id INT NOT NULL,
    location_id INT NOT NULL,
    start_date DATE NOT NULL,
    end_date DATE,
    UNIQUE (family_member_id, location_id, start_date),
    FOREIGN KEY (family_member_id) REFERENCES FamilyMember(family_member_id),
    FOREIGN KEY (location_id) REFERENCES Location(location_id),
    CHECK (end_date IS NULL OR end_date >= start_date)
);

CREATE TABLE ClubMember (
    membership_number INT AUTO_INCREMENT PRIMARY KEY,
    location_id INT NOT NULL,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    date_of_birth DATE NOT NULL,
    registration_date DATE NOT NULL,
    height_cm DECIMAL(5,2),
    weight_kg DECIMAL(5,2),
    ssn VARCHAR(15) UNIQUE,
    medicare_number VARCHAR(20) UNIQUE,
    phone_number VARCHAR(20),
    address VARCHAR(150),
    city VARCHAR(60),
    province CHAR(2),
    postal_code VARCHAR(10),
    FOREIGN KEY (location_id) REFERENCES Location(location_id)
);

CREATE TABLE ClubMemberFamilyRelation (
    relation_id INT AUTO_INCREMENT PRIMARY KEY,
    membership_number INT NOT NULL,
    family_member_id INT NOT NULL,
    relationship_type ENUM(
        'Father', 'Mother', 'Grandfather', 'Grandmother',
        'Tutor', 'Partner', 'Friend', 'Other'
    ) NOT NULL,
    start_date DATE NOT NULL,
    end_date DATE,
    UNIQUE (membership_number, family_member_id, start_date),
    FOREIGN KEY (membership_number)
        REFERENCES ClubMember(membership_number),
    FOREIGN KEY (family_member_id)
        REFERENCES FamilyMember(family_member_id),
    CHECK (end_date IS NULL OR end_date >= start_date)
);

CREATE TABLE Hobby (
    hobby_id INT AUTO_INCREMENT PRIMARY KEY,
    hobby_name VARCHAR(50) NOT NULL UNIQUE
);

CREATE TABLE ClubMemberHobby (
    membership_number INT NOT NULL,
    hobby_id INT NOT NULL,
    PRIMARY KEY (membership_number, hobby_id),
    FOREIGN KEY (membership_number)
        REFERENCES ClubMember(membership_number)
        ON DELETE CASCADE,

```

```

FOREIGN KEY (hobby_id)
  REFERENCES Hobby(hobby_id)
);

CREATE TABLE Payment (
  payment_id INT AUTO_INCREMENT PRIMARY KEY,
  membership_number INT NOT NULL,
  payment_date DATE NOT NULL,
  amount DECIMAL(10,2) NOT NULL,
  payment_method ENUM('Cash', 'Debit', 'Credit') NOT NULL,
  membership_year YEAR NOT NULL,
  installment_number INT NOT NULL DEFAULT 1,
  UNIQUE (membership_number, membership_year, installment_number),
  FOREIGN KEY (membership_number)
    REFERENCES ClubMember(membership_number),
  CHECK (amount > 0),
  CHECK (installment_number BETWEEN 1 AND 4)
);

CREATE TABLE FIFAGame (
  game_id INT AUTO_INCREMENT PRIMARY KEY,
  location_id INT NOT NULL,
  team_name VARCHAR(100) NOT NULL,
  opponent_name VARCHAR(100) NOT NULL,
  game_date DATE NOT NULL,
  team_score INT NOT NULL,
  opponent_score INT NOT NULL,
  FOREIGN KEY (location_id) REFERENCES Location(location_id)
);

CREATE TABLE FIFAParticipation (
  game_id INT NOT NULL,
  membership_number INT NOT NULL,
  PRIMARY KEY (game_id, membership_number),
  FOREIGN KEY (game_id)
    REFERENCES FIFAGame(game_id)
    ON DELETE CASCADE,
  FOREIGN KEY (membership_number)
    REFERENCES ClubMember(membership_number)
);

```

3. SQL Queries and Transactions

The eight required queries below are implemented in `sql/03_queries.sql`. Each corresponds to one item (i–viii) from the project description.

```
-- Query (i): Complete details for every location

WITH Phones AS (
    SELECT location_id, GROUP_CONCAT(phone_number SEPARATOR ', ') AS phone_numbers
    FROM LocationPhone
    GROUP BY location_id
),
GM AS (
    SELECT pa.location_id,
           CONCAT(p.first_name, ' ', p.last_name) AS gm_name
    FROM PersonnelAssignment pa
    JOIN Personnel p ON p.personnel_id = pa.personnel_id
    WHERE p.role = 'General Manager'
           AND pa.end_date IS NULL
),
PersonnelCount AS (
    SELECT location_id, COUNT(DISTINCT personnel_id) AS num_personnel
    FROM PersonnelAssignment
    WHERE end_date IS NULL
    GROUP BY location_id
),
MemberCount AS (
    SELECT location_id, COUNT(*) AS num_club_members
    FROM ClubMember
    GROUP BY location_id
),
FifaMemberCount AS (
    SELECT cm.location_id, COUNT(DISTINCT fp.membership_number) AS num_fifa_players
    FROM ClubMember cm
    JOIN FIFAParticipation fp ON fp.membership_number = cm.membership_number
    GROUP BY cm.location_id
)
SELECT
    l.name           AS location_name,
    l.address,
    l.city,
    l.province,
    l.postal_code,
    ph.phone_numbers,
    l.web_address,
    l.location_type,
    l.capacity,
    gm.gm_name       AS general_manager,
    COALESCE(pc.num_personnel, 0) AS num_personnel,
    COALESCE(mc.num_club_members, 0) AS num_club_members,
    COALESCE(fc.num_fifa_players, 0) AS num_fifa_players
FROM Location l
LEFT JOIN Phones ph      ON ph.location_id = l.location_id
LEFT JOIN GM gm         ON gm.location_id = l.location_id
LEFT JOIN PersonnelCount pc ON pc.location_id = l.location_id
LEFT JOIN MemberCount mc  ON mc.location_id = l.location_id
LEFT JOIN FifaMemberCount fc ON fc.location_id = l.location_id
ORDER BY num_club_members ASC;

-- Query (ii): Major club members who played >= 1 FIFA game

WITH AgeCalc AS (
    SELECT membership_number,
           TIMESTAMPDIFF(YEAR, date_of_birth, CURDATE()) AS age
    FROM ClubMember
),
PrevYearPay AS (
    SELECT membership_number, SUM(amount) AS total_paid
    FROM Payment
    WHERE membership_year = YEAR(CURDATE()) - 1
    GROUP BY membership_number
),
GameCounts AS (
```



```

SELECT membership_number, COUNT(*) AS num_games
FROM FIFAParticipation
GROUP BY membership_number
)
SELECT
    l.name AS location_name,
    cm.membership_number,
    cm.first_name,
    cm.last_name,
    a.age,
    cm.city,
    cm.province,
    CASE WHEN COALESCE(p.total_paid, 0) >= 200 THEN 'Active' ELSE 'Inactive' END AS status,
    g.num_games
FROM ClubMember cm
JOIN AgeCalc a      ON a.membership_number = cm.membership_number
JOIN Location l     ON l.location_id = cm.location_id
JOIN GameCounts g   ON g.membership_number = cm.membership_number
LEFT JOIN PrevYearPay p ON p.membership_number = cm.membership_number
WHERE a.age >= 18
ORDER BY g.num_games ASC;

```

-- Query (iii): Club members with >= 4 different hobbies

```

WITH AgeCalc AS (
    SELECT membership_number,
           TIMESTAMPDIFF(YEAR, date_of_birth, CURDATE()) AS age
    FROM ClubMember
),
PrevYearPay AS (
    SELECT membership_number, SUM(amount) AS total_paid
    FROM Payment
    WHERE membership_year = YEAR(CURDATE()) - 1
    GROUP BY membership_number
),
HobbyCounts AS (
    SELECT membership_number, COUNT(DISTINCT hobby_id) AS num_hobbies
    FROM ClubMemberHobby
    GROUP BY membership_number
    HAVING COUNT(DISTINCT hobby_id) >= 4
)
SELECT
    l.name AS location_name,
    cm.membership_number,
    cm.first_name,
    cm.last_name,
    a.age,
    cm.city,
    cm.province,
    CASE
        WHEN COALESCE(p.total_paid, 0) >= (CASE WHEN a.age >= 18 THEN 200 ELSE 100 END)
        THEN 'Active' ELSE 'Inactive'
    END AS status,
    h.num_hobbies
FROM ClubMember cm
JOIN AgeCalc a      ON a.membership_number = cm.membership_number
JOIN Location l     ON l.location_id = cm.location_id
JOIN HobbyCounts h   ON h.membership_number = cm.membership_number
LEFT JOIN PrevYearPay p ON p.membership_number = cm.membership_number
ORDER BY a.age DESC, l.name ASC;

```

-- QUERY-4 (iv) Get all major club members (18+) who have never participated in any FIFA game --

```

SELECT
    l.name AS location_name,
    cm.membership_number,
    cm.first_name,
    cm.last_name,
    TIMESTAMPDIFF(YEAR, cm.date_of_birth, CURDATE()) AS age,
    cm.city,
    cm.province,

    CASE
        WHEN COALESCE((
            SELECT SUM(p.amount)

```

```

        FROM Payment p
        WHERE p.membership_number = cm.membership_number
              AND p.membership_year = YEAR(CURDATE()) - 1
    ), 0) >=
CASE
    WHEN TIMESTAMPDIFF(
        YEAR,
        cm.date_of_birth,
        STR_TO_DATE(
            CONCAT(YEAR(CURDATE()) - 1, '-12-31'),
            '%Y-%m-%d'
        )
    ) >= 18
    THEN 200
    ELSE 100
END
    THEN 'Active'
    ELSE 'Inactive'
END AS status
FROM ClubMember cm
JOIN Location l
    ON l.location_id = cm.location_id
WHERE TIMESTAMPDIFF(
    YEAR,
    cm.date_of_birth,
    CURDATE()
) >= 18
AND NOT EXISTS (
    SELECT 1
    FROM FIFAParticipation fp
    WHERE fp.membership_number = cm.membership_number
)
ORDER BY
    location_name ASC,
    age ASC;

-- QUERY-5 (v): total number of club members for every age
SELECT
    TIMESTAMPDIFF(YEAR, date_of_birth, CURDATE()) AS age,
    COUNT(*) AS total_club_members
FROM ClubMember
GROUP BY age
ORDER BY age DESC;

-- QUERY-6 (vi): Major members who are also family members
SELECT
    cm.membership_number                AS member_number,
    CONCAT(cm.first_name, ' ', cm.last_name) AS member_name,
    child.membership_number              AS associated_member_number,
    CONCAT(child.first_name, ' ', child.last_name) AS associated_member_name,
    child.date_of_birth                  AS associated_dob,
    child.ssn                            AS associated_ssn,
    child.medicare_number                AS associated_medicare_number,
    child.phone_number                   AS associated_phone,
    child.address                        AS associated_address,
    child.city                           AS associated_city,
    child.province                       AS associated_province,
    child.postal_code                    AS associated_postal_code,
    r.relationship_type,
    CASE WHEN r.end_date IS NULL THEN 'Active' ELSE 'Ended' END AS status
FROM ClubMember cm
JOIN FamilyMember fm ON fm.ssn = cm.ssn
JOIN ClubMemberFamilyRelation r ON r.family_member_id = fm.family_member_id
JOIN ClubMember child ON child.membership_number = r.membership_number
WHERE TIMESTAMPDIFF(YEAR, cm.date_of_birth, CURDATE()) >= 18;

-- QUERY-7 (vii): Fees, major members, 2023-2025
SELECT
    cm.membership_number,
    CONCAT(cm.first_name, ' ', cm.last_name) AS member_name,
    p.membership_year,
    SUM(p.amount) AS total_paid
FROM ClubMember cm
JOIN Payment p ON p.membership_number = cm.membership_number
WHERE TIMESTAMPDIFF(YEAR, cm.date_of_birth, CURDATE()) >= 18

```

```
    AND p.membership_year BETWEEN 2023 AND 2025
GROUP BY cm.membership_number, member_name, p.membership_year
ORDER BY cm.membership_number, p.membership_year;

-- QUERY-8 (viii): Members with >=4 FIFA games (all members)
SELECT
    cm.membership_number,
    CONCAT(cm.first_name, ' ', cm.last_name) AS member_name,
    COUNT(fp.game_id) AS games_played
FROM ClubMember cm
JOIN FIFAParticipation fp ON fp.membership_number = cm.membership_number
GROUP BY cm.membership_number, member_name
HAVING COUNT(fp.game_id) >= 4;
```

4. Database Population

The database is populated using `sql/02_seed.sql`, inserted in an order that respects foreign-key dependencies (Location before ClubMember, ClubMember before Payment, etc.). The specification requires at least 10 representative tuples per table such that every query returns at least two rows.

4.1 Current Row Counts

Table	Row Count	Meets ≥ 10 requirement?
Location	10	Yes
LocationPhone	11	Yes
Personnel	10	Yes
PersonnelAssignment	10	Yes
FamilyMember	10	Yes
FamilyMemberAssignment	10	Yes
ClubMember	10	Yes
ClubMemberFamilyRelation	10	Yes
Hobby	10	Yes
ClubMemberHobby	18	Yes
Payment	18	Yes
FIFAGame	10	Yes
FIFAParticipation	29	Yes

4.2 Seed Data (02_seed.sql)

```
-- =====
-- CSCS (Country Soccer Club System) - Seed Data
-- Insert order respects FK dependencies:
-- Location -> LocationPhone
-- Personnel -> PersonnelAssignment
-- FamilyMember -> FamilyMemberAssignment
-- Location -> ClubMember -> ClubMemberFamilyRelation
-- Hobby -> ClubMemberHobby
-- ClubMember -> Payment
-- Location -> FIFAGame -> FIFAParticipation
-- =====

USE wqc353_1;

-- =====
-- Table: Location
-- =====
INSERT INTO Location (location_id, location_type, name, address, city, province, postal_code, web_address, capacity) VALUES
(1, 'Head', 'CSCS Head Office', '100 Rue Sainte-Catherine', 'Montreal', 'QC', 'H2X1K4', 'https://cscs.ca', 500),
(2, 'Branch', 'CSCS West Branch', '450 Boulevard Cur -Labelle', 'Laval', 'QC', 'H7P2P2', 'https://west.cscs.ca', 200),
(3, 'Branch', 'CSCS East Branch', '25 Rue Saint-Charles', 'Longueuil', 'QC', 'J4H1E5', 'https://east.cscs.ca', 150),
(4, 'Branch', 'CSCS North Branch', '200 Boulevard Laurentien', 'Saint-Laurent', 'QC', 'H4M2M2', 'https://north.cscs.ca', 175),
(5, 'Branch', 'CSCS South Branch', '300 Rue Taschereau', 'Brossard', 'QC', 'J4Y1A1', 'https://south.cscs.ca', 180),
(6, 'Branch', 'CSCS Central Branch', '500 Rue Sainte-Anne', 'Varenes', 'QC', 'J3X1P7', 'https://central.cscs.ca', 160),
(7, 'Branch', 'CSCS Plateau Branch', '80 Avenue du Mont-Royal E', 'Montreal', 'QC', 'H2T1P1', 'https://plateau.cscs.ca', 120),
(8, 'Branch', 'CSCS Verdun Branch', '5 Rue de Verdun', 'Montreal', 'QC', 'H4G2M5', 'https://verdun.cscs.ca', 130),
(9, 'Branch', 'CSCS Rive-Nord Branch', '750 Chemin du Bas-de-la-Riviere', 'Terrebonne', 'QC', 'J6W1A3', 'https://rivenord.cscs.ca', 140),
(10, 'Branch', 'CSCS Anjou Branch', '7000 Rue Sherbrooke E', 'Montreal', 'QC', 'H1K1C5', 'https://anjou.cscs.ca', 110);

-- =====
-- Table: LocationPhone
-- =====
INSERT INTO LocationPhone (location_id, phone_number) VALUES
(1, '514-555-0100'),
(1, '514-555-0101'),
(2, '450-555-0200'),
(3, '450-555-0300'),
(4, '450-555-0400'),
(5, '450-555-0500'),
(6, '450-555-0600'),
(7, '514-555-0700'),
(8, '514-555-0800'),
(9, '450-555-0900'),
(10, '514-566-0120');

-- =====
-- Table: Personnel
-- =====
```

```

INSERT INTO Personnel (personnel_id, first_name, last_name, date_of_birth, ssn, medicare_number, phone_number, address, city, province,
postal_code, email, role, mandate) VALUES
(1, 'Marc', 'Tremblay', '1975-03-12', '111-111-111', 'TREM75031299', '514-555-1001', '12 Rue Principale', 'Montreal', 'QC', 'H1A1A1',
'marc.tremblay@cscs.ca', 'General Manager', 'Salaried'),
(2, 'Julie', 'Bouchard', '1982-07-04', '222-222-222', 'BOUC82070499', '514-555-1002', '34 Avenue du Parc', 'Montreal', 'QC', 'H2B2B2',
'julie.bouchard@cscs.ca', 'Treasurer', 'Salaried'),
(3, 'Simon', 'Roy', '1990-11-23', '333-333-333', 'ROYS90112399', '450-555-1003', '56 Rue des Érables', 'Laval', 'QC', 'H7C3C3',
'simon.roy@cscs.ca', 'Coach', 'Volunteer'),
(4, 'Nadia', 'Cormier', '1988-01-30', '444-444-444', 'CORM88013099', '450-555-1004', '78 Rue de la Gare', 'Longueuil', 'QC', 'J4D4D4',
'nadia.cormier@cscs.ca', 'Secretary', 'Volunteer'),
(5, 'Eric', 'Fortin', '1995-05-17', '555-555-555', 'FORT95051799', '514-555-1005', '90 Rue Saint-Denis', 'Montreal', 'QC', 'H2E5E5',
'eric.fortin@cscs.ca', 'Assistant Coach', 'Volunteer'),
(6, 'Luc', 'Gauthier', '1985-06-14', '601-111-111', 'GAUT85061499', '514-555-1006', '22 Rue Masson', 'Montreal', 'QC', 'H2S2S2',
'luc.gauthier@cscs.ca', 'Deputy Manager', 'Salaried'),
(7, 'Isabeau', 'Pelletier', '1992-08-20', '602-111-111', 'PELL92082099', '514-555-1007', '63 Rue Beaubien', 'Montreal', 'QC', 'H2G3G3',
'isabeau.pelletier@cscs.ca', 'Captain', 'Volunteer'),
(8, 'Francois', 'Lapointe', '1987-04-05', '603-111-111', 'LAP087040599', '450-555-1008', '15 Rue Joliette', 'Brossard', 'QC', 'J4X4X4',
'francois.lapointe@cscs.ca', 'Administrator', 'Salaried'),
(9, 'Marie-Eve', 'Ouellet', '1993-12-11', '604-111-111', 'OUEL93121199', '450-555-1009', '88 Rue des Saules', 'Terrebonne', 'QC', 'J6W2W2',
'marieeve.ouellet@cscs.ca', 'Coach', 'Salaried'),
(10, 'David', 'Cloutier', '1980-09-03', '605-111-111', 'CLOU80090399', '514-555-1010', '44 Rue Notre-Dame O', 'Montreal', 'QC', 'H3C3C3',
'david.cloutier@cscs.ca', 'Secretary', 'Volunteer');

-- =====
-- Table: PersonnelAssignment
-- =====
INSERT INTO PersonnelAssignment (assignment_id, personnel_id, location_id, start_date, end_date) VALUES
(1, 1, 1, '2019-01-01', NULL), -- current GM at Head office
(2, 2, 1, '2020-06-01', NULL), -- current Treasurer at Head office
(3, 3, 2, '2021-03-15', NULL), -- current Coach at West branch
(4, 4, 3, '2022-02-01', NULL), -- current Secretary at East branch
(5, 5, 2, '2018-09-01', '2021-03-01'), -- past Assistant Coach assignment at West branch
(6, 5, 3, '2021-03-16', NULL), -- same person, now current at East branch
(7, 6, 1, '2022-01-10', NULL), -- Luc: Deputy Manager at Head office
(8, 7, 4, '2023-03-01', NULL), -- Isabeau: Captain at North branch
(9, 8, 5, '2023-07-15', NULL), -- Francois: Administrator at South branch
(10, 9, 6, '2024-01-01', NULL); -- Marie-Eve: Coach at Central branch

-- =====
-- Table: FamilyMember
-- =====
INSERT INTO FamilyMember (family_member_id, first_name, last_name, date_of_birth, ssn, medicare_number, phone_number, address, city,
province, postal_code, email) VALUES
(1, 'Denise', 'Lavoie', '1970-04-08', '666-666-666', 'LAVO70040899', '514-555-2001', '15 Rue Sherbrooke', 'Montreal', 'QC', 'H2F6F6',
'denise.lavoie@email.com'),
(2, 'Patrick', 'Girard', '1972-09-19', '777-777-777', 'GIRA72091999', '450-555-2002', '27 Rue Lévis', 'Laval', 'QC', 'H7G7G7',
'patrick.girard@email.com'),
(3, 'Chantal', 'Bergeron', '1980-12-01', '888-888-888', 'BERG80120199', '450-555-2003', '39 Rue Fournier', 'Longueuil', 'QC', 'J4H8H8',
'chantal.bergeron@email.com'),
(4, 'Alain', 'Morin', '1978-06-25', '999-999-999', 'MORI78062599', '514-555-2004', '41 Rue Rachel', 'Montreal', 'QC', 'H2I9I9',
'alain.morin@email.com'),
(5, 'Robert', 'Tremblay', '1965-07-14', '101-010-101', 'TREM65071499', '514-555-2005', '72 Rue Beaubien', 'Montreal', 'QC', 'H2G3G3',
'robert.tremblay@email.com'),
(6, 'Martine', 'Gagnon', '1973-02-28', '202-020-202', 'GAGN73022899', '450-555-2006', '100 Rue du Boise', 'Laval', 'QC', 'H7R5R5',
'martine.gagnon@email.com'),
(7, 'Louis', 'Bedard', '1968-11-09', '303-030-303', 'BEDA68110999', '450-555-2007', '55 Rue des Tilleuls', 'Longueuil', 'QC', 'J4L7L7',
'louis.bedard@email.com'),
(8, 'Sylvie', 'Charron', '1975-05-03', '404-040-404', 'CHAR75050399', '514-555-2008', '18 Cote-des-Neiges', 'Montreal', 'QC', 'H3H1A1',
'sylvie.charron@email.com'),
(9, 'Pierre', 'Beaumont', '1969-08-16', '505-050-505', 'BEAU69081699', '450-555-2009', '33 Rue Edouard', 'Brossard', 'QC', 'J4Y2Y2',
'pierre.beaumont@email.com'),
(10, 'Helene', 'Leblanc', '1977-03-22', '606-060-606', 'LEBL77032299', '514-555-2010', '90 Boulevard Decarie', 'Montreal', 'QC', 'H4L3L3',
'helene.leblanc@email.com');

-- =====
-- Table: FamilyMemberAssignment
-- =====
INSERT INTO FamilyMemberAssignment (assignment_id, family_member_id, location_id, start_date, end_date) VALUES
(1, 1, 1, '2020-01-10', NULL),
(2, 2, 2, '2021-04-01', NULL),
(3, 3, 3, '2022-05-15', NULL),
(4, 4, 1, '2019-08-01', '2023-01-01'),
(5, 5, 4, '2023-03-01', NULL),
(6, 6, 2, '2021-09-15', NULL),
(7, 7, 3, '2022-11-01', NULL),
(8, 8, 5, '2023-08-20', NULL),
(9, 9, 1, '2024-01-05', NULL),
(10, 10, 6, '2024-03-10', NULL);

-- =====
-- Table: ClubMember
-- =====

```

```

INSERT INTO ClubMember (membership_number, location_id, first_name, last_name, date_of_birth, registration_date, height_cm, weight_kg, ssn,
medicare_number, phone_number, address, city, province, postal_code) VALUES
(1, 1, 'Thomas', 'Lavoie', '2010-02-14', '2022-08-01', 155.00, 45.50, 'M100000001', 'LAV010021499', '514-555-3001', '15 Rue Sherbrooke',
'Montreal', 'QC', 'H2F6F6'),
(2, 1, 'Sophie', 'Bélanger', '2008-06-30', '2021-09-05', 162.00, 52.00, 'M100000002', 'BELA08063099', '514-555-3002', '20 Rue Ontario',
'Montreal', 'QC', 'H2K1K1'),
(3, 2, 'Maxime', 'Girard', '2011-11-02', '2023-01-15', 148.00, 40.00, 'M100000003', 'GIRA11110299', '450-555-3003', '27 Rue Lévis', 'Laval',
'QC', 'H7G7G7'),
(4, 2, 'Camille', 'Dubois', '2005-03-19', '2020-02-20', 170.00, 60.00, 'M100000004', 'DUBO05031999', '450-555-3004', '10 Rue des Pins',
'Laval', 'QC', 'H7L2L2'),
(5, 3, 'Antoine', 'Bergeron', '2012-09-08', '2023-06-10', 140.00, 36.00, 'M100000005', 'BERG12090899', '450-555-3005', '39 Rue Fournier',
'Longueuil', 'QC', 'J4H8H8'),
(6, 3, 'Laurie', 'Pelletier', '2007-12-25', '2021-11-01', 158.00, 48.00, 'M100000006', 'PELL07122599', '450-555-3006', '5 Rue Curé-
Poirier', 'Longueuil', 'QC', 'J4J3J3'),
(7, 1, 'Gabriel', 'Morin', '2006-04-17', '2020-09-01', 168.00, 58.00, 'M100000007', 'MORI06041799', '514-555-3007', '41 Rue Rachel',
'Montreal', 'QC', 'H2I9I9'),
(8, 2, 'Emma', 'Roy', '2013-01-05', '2024-01-20', 135.00, 32.00, 'M100000008', 'ROYE13010599', '450-555-3008', '56 Rue des Érables', 'Laval',
'QC', 'H7C3C3'),
(9, 1, 'Alexandre', 'Gagnon', '1998-08-14', '2023-05-10', 181.00, 76.00, 'M100000009', 'GAGN98081499', '514-555-3009', '72 Rue Saint-
Laurent', 'Montreal', 'QC', 'H2Y2Y2'),
(10, 3, 'Isabelle', 'Caron', '2001-10-22', '2024-02-15', 168.00, 59.00, 'M100000010', 'CARO01102299', '450-555-3010', '88 Rue Victoria',
'Longueuil', 'QC', 'J4K4K4');

-- =====
-- Table: ClubMemberFamilyRelation
-- =====
INSERT INTO ClubMemberFamilyRelation (relation_id, membership_number, family_member_id, relationship_type, start_date, end_date) VALUES
(1, 1, 1, 'Mother', '2022-08-01', NULL),
(2, 3, 2, 'Father', '2023-01-15', NULL),
(3, 5, 3, 'Mother', '2023-06-10', NULL),
(4, 7, 4, 'Father', '2020-09-01', NULL),
(5, 2, 5, 'Father', '2021-09-05', NULL),
(6, 4, 6, 'Mother', '2020-02-20', NULL),
(7, 6, 7, 'Father', '2021-11-01', NULL),
(8, 8, 8, 'Mother', '2024-01-20', NULL),
(9, 9, 9, 'Father', '2023-05-10', NULL),
(10, 10, 10, 'Mother', '2024-02-15', NULL);

-- =====
-- Table: Hobby
-- =====
INSERT INTO Hobby (hobby_id, hobby_name) VALUES
(1, 'Reading'),
(2, 'Swimming'),
(3, 'Video Games'),
(4, 'Music'),
(5, 'Chess'),
(6, 'Painting'),
(7, 'Cycling'),
(8, 'Cooking'),
(9, 'Photography'),
(10, 'Dancing');

-- =====
-- Table: ClubMemberHobby
-- =====
INSERT INTO ClubMemberHobby (membership_number, hobby_id) VALUES
(1, 2), (1, 5), -- Thomas: Swimming, Chess
(2, 1), (2, 4), -- Sophie: Reading, Music
(3, 3), -- Maxime: Video Games
(4, 1), (4, 2), (4, 4), -- Camille: Reading, Swimming, Music
(5, 5), -- Antoine: Chess
(6, 3), (6, 1), -- Laurie: Video Games, Reading
(7, 2), -- Gabriel: Swimming
(8, 4); -- Emma: Music

-- =====
-- Table: Payment
-- =====
INSERT INTO Payment (payment_id, membership_number, payment_date, amount, payment_method, membership_year, installment_number) VALUES
-- Thomas: paid in full
(1, 1, '2024-08-01', 300.00, 'Debit', 2024, 1),
-- Sophie: paid in 2 installments
(2, 2, '2024-09-01', 150.00, 'Credit', 2024, 1),
(3, 2, '2024-11-01', 150.00, 'Credit', 2024, 2),
-- Maxime: paid in full
(4, 3, '2024-01-15', 300.00, 'Cash', 2024, 1),
-- Camille: paid in 4 installments
(5, 4, '2024-02-20', 75.00, 'Debit', 2024, 1),
(6, 4, '2024-05-20', 75.00, 'Debit', 2024, 2),
(7, 4, '2024-08-20', 75.00, 'Debit', 2024, 3),
(8, 4, '2024-11-20', 75.00, 'Debit', 2024, 4),

```

```

-- Antoine: paid in full
(9, 5, '2024-06-10', 300.00, 'Cash', 2024, 1),
-- Laurie: paid in 2 installments
(10, 6, '2024-11-01', 150.00, 'Credit', 2024, 1),
(11, 6, '2025-01-01', 150.00, 'Credit', 2024, 2),
-- Gabriel: paid in full
(12, 7, '2024-09-01', 300.00, 'Debit', 2024, 1),
-- Emma: only first installment so far (still owes the rest)
(13, 8, '2024-01-20', 75.00, 'Cash', 2024, 1);

-- =====
-- Table: FIFAGame
-- =====
INSERT INTO FIFAGame (game_id, location_id, team_name, opponent_name, game_date, team_score, opponent_score) VALUES
(1, 1, 'CSCS Head Office', 'Montreal United', '2024-03-10', 3, 1),
(2, 2, 'CSCS West Branch', 'Laval Rovers', '2024-04-05', 2, 2),
(3, 3, 'CSCS East Branch', 'Longueuil FC', '2024-05-12', 1, 4),
(4, 1, 'CSCS Head Office', 'South Shore Kickers', '2024-06-20', 5, 0),
(5, 4, 'CSCS North Branch', 'Pointe-Claire FC', '2024-07-14', 2, 1),
(6, 5, 'CSCS South Branch', 'Brossard United', '2024-08-03', 0, 3),
(7, 2, 'CSCS West Branch', 'Laval Stars', '2024-09-21', 4, 2),
(8, 1, 'CSCS Head Office', 'Outremont Athletic', '2024-10-05', 1, 1),
(9, 3, 'CSCS East Branch', 'Greenfield Park SC', '2024-11-02', 3, 0),
(10, 6, 'CSCS Central Branch', 'Varenes City FC', '2024-12-07', 2, 2);

-- =====
-- Table: FIFAParticipation
-- =====
INSERT INTO FIFAParticipation (game_id, membership_number) VALUES
(1, 1), (1, 2), (1, 7), -- Game 1: Thomas, Sophie, Gabriel
(2, 3), (2, 4), (2, 8), -- Game 2: Maxime, Camille, Emma
(3, 5), (3, 6), -- Game 3: Antoine, Laurie
(4, 1), (4, 2), (4, 7), -- Game 4: Thomas, Sophie, Gabriel again (repeat participants)
(5, 3), (5, 4), (5, 8), -- Game 5: Maxime, Camille, Emma
(6, 5), (6, 6), (6, 9), -- Game 6: Antoine, Laurie, Alexandre
(7, 1), (7, 3), (7, 4), -- Game 7: Thomas, Maxime, Camille
(8, 2), (8, 6), (8, 10), -- Game 8: Sophie, Laurie, Isabelle
(9, 7), (9, 8), (9, 9), -- Game 9: Gabriel, Emma, Alexandre
(10, 1), (10, 5), (10, 10); -- Game 10: Thomas, Antoine, Isabelle

-- =====
-- Additional seed data for testing Queries 3 and 4
-- =====

-- =====
-- Query 3 test data:
-- Club Members with at least 4 hobbies
-- =====

-- Camille currently has hobbies 1, 2, and 4.
-- Add hobbies 3 and 5, giving her 5 hobbies.
INSERT INTO ClubMemberHobby (membership_number, hobby_id) VALUES
(4, 3),
(4, 5);

-- Laurie currently has hobbies 1 and 3.
-- Add hobbies 2, 4, and 5, giving her 5 hobbies.
INSERT INTO ClubMemberHobby (membership_number, hobby_id) VALUES
(6, 2),
(6, 4),
(6, 5);

-- =====
-- Query 4 test data:
-- =====
INSERT INTO Payment (
payment_id,
membership_number,
payment_date,
amount,
payment_method,
membership_year,
installment_number
) VALUES

-- Camille: major member, owes $200, fully paid => Active
(14, 4, '2025-02-20', 100.00, 'Debit', 2025, 1),
(15, 4, '2025-06-20', 100.00, 'Debit', 2025, 2),

-- Laurie: major by the end of 2025, owes $200,
-- but paid only $100 => Inactive
(16, 6, '2025-03-01', 100.00, 'Credit', 2025, 1),

```

```
-- Alexandre: major member, fully paid $200 => Active  
(17, 9, '2025-05-10', 200.00, 'Debit', 2025, 1),  
  
-- Isabelle: major member, paid only $100 => Inactive  
(18, 10, '2025-04-15', 100.00, 'Cash', 2025, 1);
```


5. Table Verification — SELECT COUNT(*) FROM R

The output of "SELECT COUNT(*) FROM R;" for each relation R that is formed in the database is listed below. This has been done using sql/04_verify.sql on the local development database (which matches the AITS wqc353_1 database schema).

Relation (R)	COUNT(*)
Location	10
LocationPhone	11
Personnel	10
PersonnelAssignment	10
FamilyMember	10
FamilyMemberAssignment	10
ClubMember	10
ClubMemberFamilyRelation	10
Hobby	10
ClubMemberHobby	18
Payment	18
FIFAGame	10
FIFAParticipation	29

Verification Script (04_verify.sql)

```
-- =====
-- CSCS (Country Soccer Club System) - Verification
-- SELECT COUNT(*) FROM R for every relation R in the database
-- =====

USE wqc353_1;

SELECT 'Location' AS table_name, COUNT(*) AS row_count FROM Location
UNION ALL SELECT 'LocationPhone', COUNT(*) FROM LocationPhone
UNION ALL SELECT 'Personnel', COUNT(*) FROM Personnel
UNION ALL SELECT 'PersonnelAssignment', COUNT(*) FROM PersonnelAssignment
UNION ALL SELECT 'FamilyMember', COUNT(*) FROM FamilyMember
UNION ALL SELECT 'FamilyMemberAssignment', COUNT(*) FROM FamilyMemberAssignment
UNION ALL SELECT 'ClubMember', COUNT(*) FROM ClubMember
UNION ALL SELECT 'ClubMemberFamilyRelation', COUNT(*) FROM ClubMemberFamilyRelation
UNION ALL SELECT 'Hobby', COUNT(*) FROM Hobby
UNION ALL SELECT 'ClubMemberHobby', COUNT(*) FROM ClubMemberHobby
UNION ALL SELECT 'Payment', COUNT(*) FROM Payment
UNION ALL SELECT 'FIFAGame', COUNT(*) FROM FIFAGame

UNION ALL SELECT 'FIFAParticipation', COUNT(*) FROM FIFAParticipation;
```